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The Essentials

Introduction and Motivation

The **k-means** algorithm is a method for **estimating discrete latent factors** in an **unsupervised** clustering context.

The Fundamental Problem

Imagine this: You're given 1,000 unlabeled animal photos and asked: "How many different types of animals are there, and which photos go together?" This is exactly the clustering problem!

K-means answers this question by making a **simplifying assumption**: each type of animal can be represented by an **average image** (prototype), and each photo is close to one of these prototypes.

Key Concept: Clustering is **unsupervised**: no labels, no teacher to say "this is a cat." The algorithm must discover the structure by itself.

Real Data Problem

Fundamental Observation:

Data generally does **not** come from a simple Gaussian process (variations around a mean prototype).

Characteristics of Real Data:

- Data distribution typically shows **non-uniformity**
- Presence of **regions of higher density**

Definition of Clustering:

Clustering is an **unsupervised** method that aims to discover **coherent groups** of data, corresponding to **density peaks** in the data.

Geometric Intuition: The Musical Chairs Game

Practical Analogy

Imagine a game of musical chairs with K chairs (clusters) and N players (data points). When the music stops:

- Each player runs to the nearest chair (**assignment**)
- Each chair is moved to the center of the players surrounding it (**update**)
- Repeat until the chairs no longer move

This is exactly k-means!

Essential Idea: K-means alternates between two simple questions: “where does each point go?” and “where are the centers?” This alternation guarantees improvement at each step.

Overview of Clustering Methods

Main Methods

There are many clustering methods, including:

Hierarchical Methods:

- Hierarchical Agglomerative Clustering (HAC)

Partitioning Methods:

- **k-means** (Lloyd’s algorithm)
- k-medoids

Spectral Methods:

- Spectral clustering

Other Approaches:

- Community detection
- High-dimensional clustering
- Self-Organizing Maps (SOM)
- Neural Gas

Historical Note: Research on clustering has been active for at least 50 years.

Hierarchical Agglomerative Clustering

Algorithm Principle

Iterative Process:

- **Initialization:** Each data point is a cluster
- **Search:** Find the closest pair of clusters
- **Merge:** Merge these two clusters
- **Iteration:** Repeat until completion

Result: Produces a **dendrogram** of the data.

Distances Between Sets (Clusters)

Possible Metrics:

- **Distance between centers** (centroid linkage)
- **Maximum distance** (complete linkage): $\max_{x \in C_1, y \in C_2} d(x, y)$
- **Minimum distance** (single linkage): $\min_{x \in C_1, y \in C_2} d(x, y)$

Limitation: The number of clusters is **unknown a priori**.

k-means Strategy

Fundamental Assumptions

The **k-means** clustering algorithm assumes:

- A **normed metric space** (Euclidean) on the data
- An **unsupervised** assignment of data to clusters
- A **hard-assignment** algorithm: assignment is **binary**, each data point is assigned to **one and only one** cluster

Formal Model

Data:

Let $X = \{x_i\}_{i \in [[N]]} \subset \Omega \subseteq \mathbb{R}^D$, and given $K \in \mathbb{N}^*$

Model Variables:

- **Binary assignment variables (latent):**

$$Z = \{z_{ik}\} \quad \text{with} \quad z_{ik} \in \{\text{false}, \text{true}\} \equiv \{0, 1\} \quad \forall i, k$$

- **Cluster representatives:**

$$M = \{\mu_k\}_{k \in [[K]]} \quad \text{with} \quad \mu_k \in \mathbb{R}^D \quad \forall k$$

Complete Parameters:

$$\theta = [M, Z]$$

k-means Loss Function

Optimization Problem

k-means seeks the following assignment:

$$\hat{\theta} = [\hat{M}, \hat{Z}] = \arg \min_{\theta=[M,Z]} L(\theta, X)$$

Loss Function:

$$L(\theta, X) = \sum_{i=1}^N \sum_{k=1}^K z_{ik} \|x_i - \mu_k\|_2^2$$

Interpretation:

- Sum of **squared distances** between each point and its cluster representative
- The smaller this sum, the better the partition

Why squared? Two reasons:

- **Mathematical:** makes the problem easier to solve
- **Geometric:** penalizes more points that are very far away

Problem Complexity

Major Difficulty: Since the assignment is binary, exact optimization is **NP-Hard**.

Practical Solution: Seek an **approximation via coordinate descent**.

Coordinate Descent Algorithm

General Principle

Coordinate descent is an **alternating minimization algorithm**.

Problem:

Let $f : \mathbb{R}^D \rightarrow \mathbb{R}$, we seek:

$$x^* = \arg \min_{x \in \mathbb{R}^D} f(x)$$

i Mathematical Details: Resolution Strategy

Definition of Partial Functions:

For $d \in [[D]]$, define $f^d : \mathbb{R}^D \times \mathbb{R} \rightarrow \mathbb{R}$:

$$f^d(x, y) = f([x(1), \dots, x(d-1), y, x(d+1), \dots, x(D)]^T)$$

Alternating Optimization:

Seek alternately the minimum along each coordinate:

$$x^{(t+1)}(d) = \arg \min_y f^d(x^{(t)}, y)$$

Principle: Fix all variables except one, optimize this variable, then move to the next.

Application to k-means

Alternating Optimization of Z and M

We alternate the optimization of Z and M .

Definitions of Partial Losses:

$$L_M(Z) = L(\theta, X)|_{M=M}$$

$$L_Z(M) = L(\theta, X)|_{Z=Z}$$

The Two Magic Equations

Optimization with Respect to M (Representatives):

i Mathematical Details: Center Calculation

Derivative Calculation:

$$\frac{\partial L_Z}{\partial \mu_k} = -2 \sum_{i=1}^N z_{ik} (x_i - \mu_k)$$

Optimality Condition:

The optimal representative M for a given assignment Z is reached when:

$$\frac{\partial L_Z}{\partial \mu_k} = 0 \quad \Rightarrow \quad \mu_k = \frac{\sum_{i=1}^N z_{ik} x_i}{\sum_{i=1}^N z_{ik}}$$

Fundamental Conclusion: Cluster representatives are their **centers of mass** (barycenters).

Optimization with Respect to Z (Assignment):

Given a set of cluster representatives M , to minimize $L_M(Z)$:

$$z_{ik} = 1 \quad \text{only when} \quad k = \arg \min_m \|x_i - \mu_m\|_2^2$$

(and $z_{ik} = 0$ otherwise)

Conclusion: $L_M(Z)$ is minimized when **each data point is assigned to its nearest cluster representative**.

Complete k-means Algorithm

Detailed Description

Input Data:

- $X = \{x_i\}_{i \in [N]}$ with $x_i \in \mathbb{R}^D$
- $K \in \mathbb{N}^*$ (number of clusters)

Algorithm Steps

Step 1: Initialization

Initialize cluster representatives $M^{(0)}$.

Step 2: Assignment (E-step like)

Given cluster representatives $M^{(t)}$, assignment $Z^{(t)}$ associates each data point with its nearest cluster representative:

$$z_{ik}^{(t)} = \begin{cases} 1 & \text{if } k = \arg \min_m \|x_i - \mu_m^{(t)}\|_2^2 \\ 0 & \text{otherwise} \end{cases}$$

Step 3: Update (M-step like)

Given assignment $Z^{(t)}$, new cluster representatives $M^{(t+1)}$ are the centers of mass of data points assigned to clusters:

$$\mu_k^{(t+1)} = \frac{\sum_{i=1}^N z_{ik}^{(t)} x_i}{\sum_{i=1}^N z_{ik}^{(t)}}$$

Step 4: Convergence

Repeat from step 2 until convergence.

Convergence Criteria

Convergence is reached when:

- The centers of mass **do not move much**: $\|\mu_k^{(t+1)} - \mu_k^{(t)}\| < \epsilon$
- Or the assignment is **stable**: $Z^{(t+1)} = Z^{(t)}$

Important Note: Step 2 is similar to an **Expectation** step, and step 3 is similar to a **Maximization** step, considering a hard assignment (reference to the EM algorithm).

Properties of the k-means Algorithm

Theoretical Guarantees

Loss Decrease:

Since it performs **alternating minimization of convex functions**, it guarantees to **decrease the loss at each iteration**:

$$L(M^{(t+1)}, Z^{(t+1)}) \leq L(M^{(t)}, Z^{(t)})$$

Finiteness:

There is a **combinatorial but finite** number of possible assignments $\rightarrow$ the number of iterations is **(large but) finite**.

Why does it converge? Each step can only improve (or leave unchanged) the criterion. Since there's a finite number of possible assignments, the algorithm must terminate.

Geometric Interpretation

Voronoi Diagram:

Step 2 is equivalent to constructing the **discrete Voronoi diagram** of X with centers M as seeds.

Definition: The Voronoi diagram partitions space into regions, where each region contains all points closer to a given center than to any other center.

Link with the EM Algorithm

Conceptual Comparison:

- **Step 2 (Assignment):** similar to an **Expectation** step (but hard)
- **Step 3 (Update):** similar to a **Maximization** step

Key Difference: k-means uses **hard assignment** while EM uses **soft assignment**.

Sensitivity to Initialization

Problem of Local Optima

Major Limitation: Since exact optimization (optimal assignment) is **NP-Hard**, the k-means algorithm reaches a **local optimum**.

Consequences:

- The quality of the result **varies with initialization**
- Different initializations can yield very different results

The Initialization Dilemma

If all chairs start in the same corner, the game will be very long and yield a poor result.
K-means is very sensitive to initialization!

Practical Strategies

Randomization:

Randomization can be useful if computation is fast $\rightarrow$ run multiple times with different initializations.

Heuristics:

Otherwise, use **heuristics for better initialization** (like k-means++).

Initialization, Links, and Relationships

k-means++: Smart Initialization

Fundamental Principle

The principle of **k-means++** is to **maximally disperse** the initial cluster representatives $M^{(0)}$ over the data.

Demonstrated Advantages:

- Produces **higher quality** clusters
- **Accelerates convergence** of k-means

Detailed k-means++ Algorithm

Step 1: First Center

Select the first cluster representative μ_1^0 **randomly** from the data X .

Step 2: Distance Calculation

For all unselected data points x_i , calculate:

$$\Delta_i = \|x_i - \mu_m^0\|_2^2$$

the distance between x_i and its nearest representative μ_m^0 among the k representatives already selected.

Step 3: Weighted Sampling

Sample the next representative μ_{k+1}^0 from X with **probability proportional** to the distribution Δ :

$$\mathbb{P}(\text{choosing } x_i) = \frac{\Delta_i}{\sum_j \Delta_j}$$

Step 4: Iteration

Repeat from step 2 until K representatives are chosen.

Algorithm Intuition

Key Idea: Points **far from already selected centers** have a higher chance of being chosen as new centers.

Consequence: Initial centers are well **distributed in the data space**, covering different density regions.

Practical Adoption

Widespread Use: This initialization strategy is used by several Data Science packages: MATLAB™, Python SciKit Learn, R, and other standard libraries.

Practical Tip: Always use k-means++ or run k-means multiple times with different random initializations and keep the best result.

Relationship Between k-means and Gaussian Models

k-means as a Special Case of GMM

Question: Why do we say that k-means considers a **Gaussian model** for clusters?

Answer:

k-means can be seen as a limiting case of GMM (Gaussian Mixture Model) where:

Model Assumptions:

- Each cluster follows a **normal distribution**
- **Spherical and identical** covariance matrices: $\Sigma_k = \sigma^2 I_D$ for all clusters
- Variance $\sigma^2 \rightarrow 0$: distributions become spikes

Consequence:

In this limit, the **soft assignment** of GMM becomes a **hard assignment**, and we recover k-means.

i Mathematical Details: Probabilistic Formulation

If we consider the probabilistic model:

$$\mathbb{P}(x|\mu_k, \sigma^2) = \frac{1}{(2\pi\sigma^2)^{D/2}} \exp\left(-\frac{\|x - \mu_k\|^2}{2\sigma^2}\right)$$

When $\sigma \rightarrow 0$, maximizing this probability is equivalent to minimizing $\|x - \mu_k\|^2$, which is exactly the k-means criterion.

k-means is a special case of GMM! When we take a mixture of Gaussians with spherical and identical covariances $\Sigma_k = \sigma^2 I$, and let $\sigma \rightarrow 0$, then soft assignment becomes hard, and we recover k-means.

Intrinsic Link with the Euclidean Metric

Why k-means Depends on the Euclidean Metric?

Analysis of the Loss Function:

$$L(\theta, X) = \sum_{i=1}^N \sum_{k=1}^K z_{ik} \|x_i - \mu_k\|_2^2$$

The norm $\|\cdot\|_2$ is the **Euclidean norm**: $\|x\|_2^2 = \sum_d x(d)^2$.

Practical Consequences

Scale Sensitivity:

k-means is **sensitive to scale**. Variables with large values dominate the distance.

Solution: Normalize the data before applying k-means:

- Center: subtract the mean
- Scale: divide by the standard deviation

Limitation to Euclidean Spaces:

k-means cannot be directly applied to **non-Euclidean** data (graphs, texts, etc.).

Alternatives: For other metrics, use variants like k-medoids that work with arbitrary distances.

Choosing the Number of Clusters K

Problem

Difficulty: The number of clusters K must be **specified a priori** in k-means.

Selection Methods

Elbow Method:

Plot the loss L as a function of K and look for an “elbow” in the curve.

Silhouette Score:

Measures clustering quality by comparing intra-cluster distance to inter-cluster distance.

For a point i :

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$$

where $a(i)$ = average distance to other points in the same cluster, $b(i)$ = average distance to points in the nearest other cluster.

Interpretation: $s(i) \approx 1$ = well classified, ≈ 0 = on the border, ≈ -1 = misclassified.

Gap Statistic:

Compares observed loss to expected loss under a reference distribution.

Information Criteria:

- BIC (Bayesian Information Criterion)
- AIC (Akaike Information Criterion)

Practical Advice: Test several values of K and use multiple criteria.

k-means Pitfalls and How to Avoid Them

Pitfall 1: Non-Spherical Clusters

Problem: k-means assumes spherical clusters. If they're elongated or oddly shaped, it fails.

Solution:

- Standardize the data (center and scale)
- Use another method (GMM, spectral clustering)
- Transform the data (PCA first)

Pitfall 2: Very Different Cluster Sizes

Problem: k-means tends to create clusters of similar size, even if reality is different.

Solution: Adjust K, or use a method that handles this (GMM).

Pitfall 3: Sensitivity to Outliers

Problem: A very distant point pulls its cluster center toward it.

Solution:

- Detect and remove outliers beforehand
- Use k-medoids (more robust)
- Standardize to reduce influence

Pitfall 4: Result Interpretation

Problem: k-means always finds K clusters, even if the data has no structure.

Solution: Always validate with objective methods (silhouette, gap statistic).

k-means Variants

k-medoids

Difference: Centers are actual data points (not averages).

Advantage: More robust to outliers, works with any distance.

Mini-batch k-means

Difference: Uses data subsamples at each iteration.

Advantage: Much faster for large datasets.

Fuzzy C-means

Difference: Soft assignment (like GMM) but with a geometric criterion.

Advantage: Border points belong partially to multiple clusters.

Constrained Clustering

Clustering can be **constrained** with:

MUST-LINK Constraints:

Two points must be in the **same cluster**.

CANNOT-LINK Constraints:

Two points must be in **different clusters**.

Practical Application: These constraints allow incorporating **prior knowledge** or **domain constraints** into the clustering process.

In Practice: The Complete Recipe

Step 1: Data Preparation

- **Clean:** handle missing values
- **Standardize:** center and scale each variable
- **Visualize** (if possible in 2D/3D) to get an idea

Step 2: Choosing K

- **Elbow method:** plot inertia vs K
- **Silhouette score:** average over all points
- **Domain knowledge:** sometimes we know how many clusters to look for

Step 3: Execution

- Initialize K centers with k-means++
- Repeat:
 - Assign each point to the nearest center
 - Recalculate centers as averages
 - If centers move little, stop
- Evaluate with silhouette score
- If in doubt, rerun with different initializations

Step 4: Validation

- **Internal:** scores (silhouette, Davies-Bouldin)
- **External:** if you have some labels, compare
- **Manual:** visualization, common sense

Comparison: k-means vs GMM vs Hierarchical Clustering

Criterion	k-means	GMM (with EM)	Hierarchical Clustering
Assignment Type	Hard	Soft	Hierarchical
Cluster Shape	Spherical (same variance)	Elliptical (different covariances)	Flexible
Number of Clusters	Must be specified	Must be specified	Determined a posteriori (dendrogram)
Complexity	$O(N \cdot K \cdot D \cdot T)$	$O(N \cdot K \cdot D^2 \cdot T)$	$O(N^2 \log N)$ or $O(N^3)$
Advantages	Fast, simple	Probabilistic model, varied cluster shapes	No need to specify K, visualization
Disadvantages	Sensitive to initialization, assumes spherical clusters	Slower, more parameters	Expensive for large data, no reassignment

Synthesis of Key Concepts

Essential Characteristics of k-means

Nature of the Method:

- k-means is part of the **unsupervised** family of data modeling techniques
- One of the **most popular** clustering techniques

Assignment:

- k-means performs **hard assignment**
- Each point belongs to **one cluster only**

Optimization:

- **Solid** but **NP-Hard** optimization criterion (loss)
- The k-means algorithm seeks an **approximate solution** via coordinate descent (alternating minimization)

Convergence:

- The loss **is not convex** → technique **sensitive to initialization**
- k-means++ is a **prior heuristic** for initialization, effective in practice

Extensions:

- Clustering can be **constrained** (MUST-LINK and CANNOT-LINK constraints)

Condensed Cheat Sheet

In 30 seconds for the oral exam:

- k-means = clustering by distance minimization
- Two steps: assignment (points → nearest center) and update (recalculate centers)
- Problems: choice of K, initialization, spherical clusters
- Solutions: k-means++, elbow method, standardization
- Relationship: special case of GMM with $\sigma \rightarrow 0$
- Applications: segmentation, grouping, dimensionality reduction

! Mathematical Concepts to Master

Algebra and Geometry:

- Euclidean norm and distance
- Center of mass (barycenter)
- Voronoi diagram
- Metric spaces

Optimization:

- Coordinate descent
- Alternating minimization
- Combinatorial optimization
- NP-Hard complexity
- Convergence of iterative algorithms
- Local vs global optima
- Convex functions

Probability and Statistics:

- Multivariate Gaussian distribution
- Maximum likelihood
- Mixture models

Points to Develop:

- Derivation of the loss function
- Calculation of $\frac{\partial L_Z}{\partial \mu_k}$ and optimality condition
- Proof that centers are barycenters
- Proof of monotonic loss decrease
- Relationship between k-means and limiting Gaussian model